$$A\underline{x} = \lambda \underline{x}$$

<u>Self-adjoint</u>

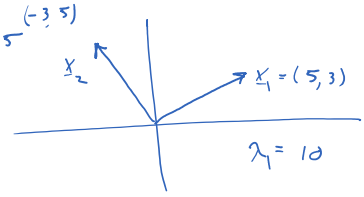
$$A = \underset{\sim}{U} \Lambda U^H \qquad U: \text{unitary}$$

$$\Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_m & & \\ & & \ddots & \\ & & & \lambda_m \end{bmatrix}$$

$$= \sum_{i=1}^{m} \lambda_i \, \underline{y}_i \underline{y}_i^H$$

want R:

$$\underline{x}_2 = \frac{1}{\sqrt{34}} \begin{bmatrix} -3 \\ 5 \end{bmatrix} \qquad \lambda_2 = 5$$



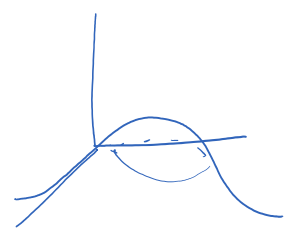
$$\underline{x}_1 = \frac{1}{\sqrt{34}} \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$R = (10)\left(\frac{1}{34}\right) \begin{bmatrix} 5 \\ 3 \end{bmatrix} \begin{bmatrix} 5 & 3 \end{bmatrix} + 5\left(\frac{1}{34}\right) \begin{bmatrix} -3 \\ 5 \end{bmatrix} \begin{bmatrix} -3 & 5 \end{bmatrix}$$

$$X \sim N(\underline{\mu}, R)$$

$$f(\underline{x}) = (k) \exp\left[\frac{-1}{2}(\underline{x}-\underline{\mu})^T R^{-1} (\underline{x}-\underline{\mu})\right]$$

want plot $\underline{x}$ s.t. $f(\underline{x}) =$ constant.
(contour plot)



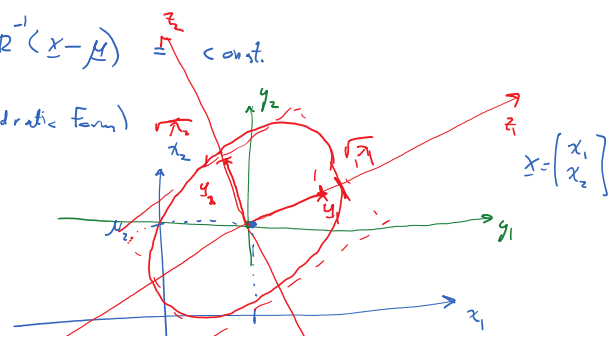
$f(\underline{x}) =$ cons.

$$\log f(x) \longrightarrow (\underline{x}-\underline{\mu})^T R^{-1} (\underline{x}-\underline{\mu}) = \text{const.}$$

(Quadratic Form)

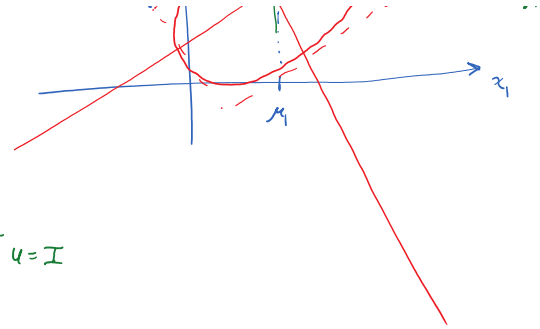
$$\underline{y} = \underline{x} - \underline{\mu}$$

$$y^T D^{-1} y$$



$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$y^T R^{-1} y$$



$$R = U \Lambda U^T$$

$$U = [\, y_1 \;\; y_2 \,] \qquad U^T U = I$$

$$\Lambda = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix}$$

$$z = U^T y \qquad y = U z$$

suppose $z = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$y = U z = y_1$

$z = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$y = U z = y_2$

$$R^{-1} = U \Lambda^{-1} U^T$$

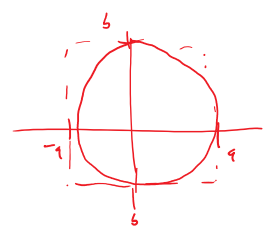
$$y^T R^{-1} y = z^T \cancel{U^T} \cancel{U} \Lambda^{-1} \cancel{U^T} \cancel{U} z$$

$$= z^T \Lambda^{-1} z$$

$$= [\, z_1 \;\; z_2 \,] \begin{bmatrix} 1/\lambda_1 & \\ & 1/\lambda_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$= \frac{z_1^2}{\lambda_1} + \frac{z_2^2}{\lambda_2} = \text{const.}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



<u>Rayleigh Quotient</u>

$$R_A(\underline{x}) = \frac{\underline{x}^H A \underline{x}}{\underline{x}^H \underline{x}} \qquad A: \text{ self adjoint}$$

$$\max_{\underline{x} \neq 0} R_A(\underline{x})$$

$$\max_{\|\underline{x}\|_2 = 1} R_A(\underline{x}) \qquad \begin{array}{ll} \max & \underline{x}^H A \underline{x} \\ \text{s.t.} & \underline{x}^H \underline{x} = 1 \end{array}$$

$$A \underline{x} = \lambda \underline{x}$$

$$(A + \sigma^2 I)\, \underline{y} = \mu \underline{y}$$

$$A \underline{y} + \sigma^2 \underline{y} = \mu \underline{y}$$

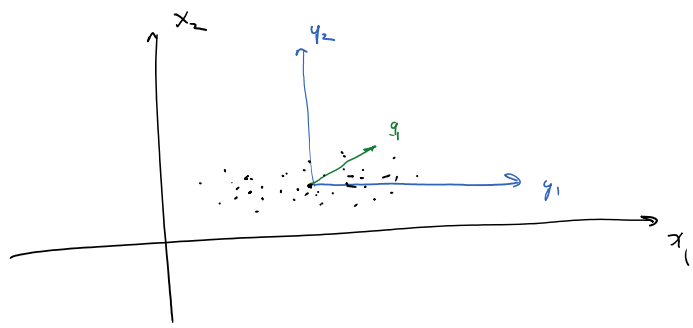
$$\boxed{A\underline{y} = (\mu - \sigma^2)\,\underline{y}}$$

$\underline{y} = \underline{x}$ (eigenvectors of $A + \sigma^2 I$ are the same as for A)

$\mu - \sigma^2 = \lambda$

$$\boxed{\mu = \lambda + \sigma^2}$$

Principal Components



Data: $\underline{x}_1, \underline{x}_2, \ldots, \underline{x}_N$, $\underline{x}_i \in \mathbb{R}^d$

Mean: $\frac{1}{N}\sum_{i=1}^{N} \underline{x}_i = \bar{\underline{x}}$

$$\underline{y}_i = \underline{x}_i - \bar{\underline{x}}$$

Direction vectors: $\underline{q}_1$: $\|\underline{q}_1\| = 1$ $\quad \underline{q}_1^T \underline{q}_1 = 1$

component of $\underline{y}_i$ in the $\underline{q}_1$ direction is

$$\langle \underline{y}_i, \underline{q}_1 \rangle = \underline{q}_1^T \underline{y}_i$$

Form $J = \frac{1}{N}\sum_{i=1}^{N} (\underline{q}_1^T \underline{y}_i)^2$

Goal: choose $\underline{q}_1$ to make J as large as possible.

Let $S = \underbrace{\frac{1}{N}\sum_{i=1}^{N} \underline{y}_i \underline{y}_i^T}$ (scatter matrix)

$$J = \frac{1}{N}\sum_{i=1}^{N} \underline{q}_1^T \underline{y}_i \underline{y}_i^T \underline{q}_1$$

$$= \underline{q}_1^T \frac{1}{N}\left[\sum_{i=1}^{N} \underline{y}_i \underline{y}_i^T\right] \underline{q}_1$$

$\max \; \underline{q}_1^T S \underline{q}_1$

s.t. $\underline{q}_1^T \ldots$

$$= q_1^T \frac{1}{N}\left[\sum_{i=1}^{N} y_i y_i^T\right] q_1$$

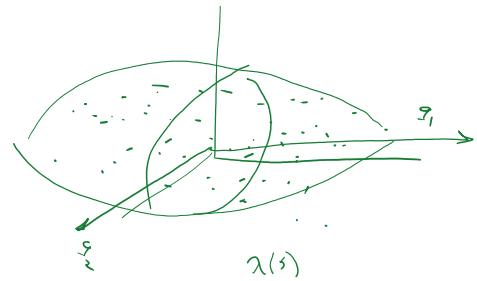
$$= q_1^T S q_1$$

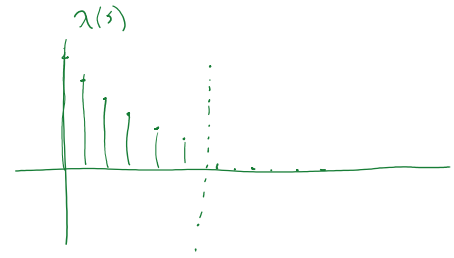
$$\left[\begin{array}{ll} \max & q_1^T S q_1 \\ \text{s.t.} & q_1^T q_1 = 1 \end{array}\right.$$

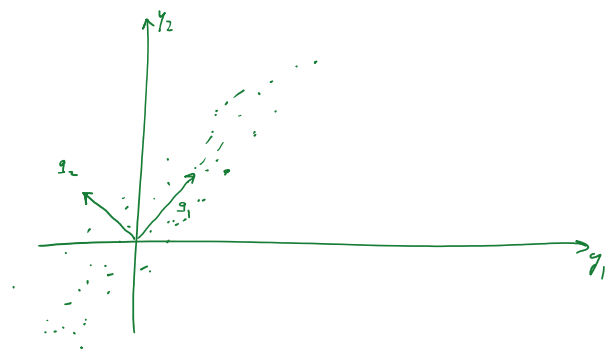
$$\max_{q_2} \quad q_2^T S q_2$$

$$\text{s.t.} \quad q_2^T q_2 = 1$$

$$q_2^T q_1 = 0$$





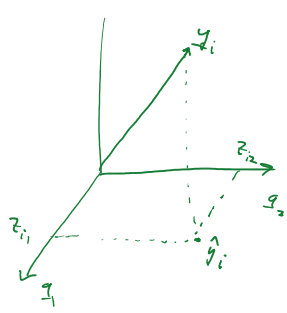


$$z_{i1} = q_1^T y_i$$

$$z_{i2} = q_2^T y_i$$

$$\vdots$$

$$z_{ip} = q_p^T y_i$$

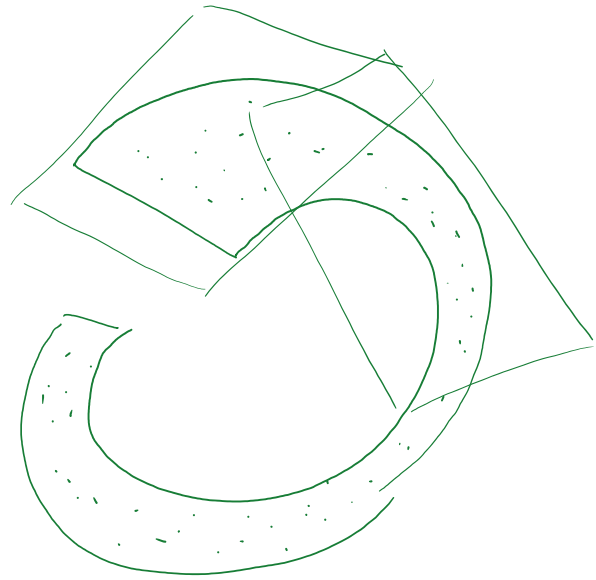


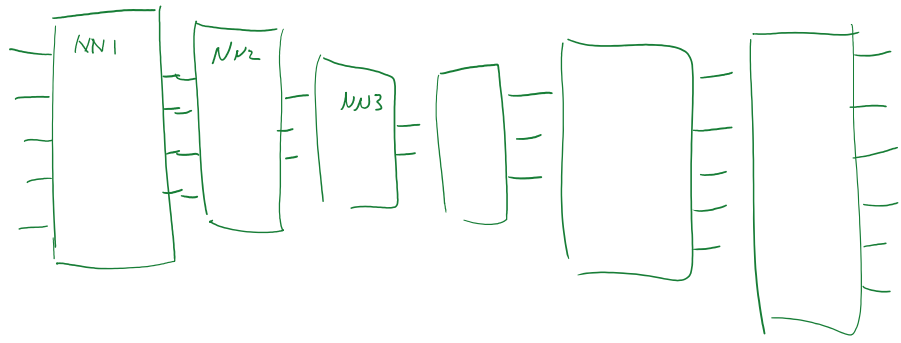
$$\begin{bmatrix} z_{i1} \\ z_{i2} \\ \vdots \\ z_{ip} \end{bmatrix}$$

$$\hat{y}_i = z_{i1} q_1 + z_{i2} q_2 + \cdots + z_{ip} q_p$$

---

v_{q_1}





$y_i \; y_i^T$

$$\begin{bmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{i_{200}} \end{bmatrix} \begin{bmatrix} y_{i1} & y_{i2} & \cdots & y_{i_{200}} \end{bmatrix}$$

Signal subspace Techniques

$x[t] = A \sin(\omega_0 t)$

$\downarrow$

$x[t+2] + a_1 x[t+1] + a_2 x[t] = 0$

Signal model:

$$x[t] = \sum_{i=1}^{p} a_i e^{j(2\pi f_i t + \phi_i)}$$

p modes

f_i = freq.

amplitudes

phase

$f_i \in [-0.5, 0.5]$

ϕ_i : phase
random $\sim \mathcal{U}(0, 2\pi)$
independent

Autocorrelation:

$$r_{xx}[k] = E[x[t]\, x[t-k]^*]$$

$$= E\left[\left(\sum_{i=1}^{P} a_i e^{j(2\pi f_i t + \phi_i)}\right)\left(\sum_{\ell=1}^{P} a_\ell^* e^{-j(2\pi f_\ell (t-k) + \phi_\ell)}\right)\right]$$

$$= \sum_{i=1}^{P}\sum_{\ell=1}^{P} a_i a_\ell^* e^{j(2\pi f_i t)} e^{-j2\pi f_\ell (t-k)} \underbrace{E\left[e^{j\phi_i} e^{-j\phi_\ell}\right]}_{E\left[e^{j(\phi_i - \phi_\ell)}\right] = \begin{cases} 1 & \text{if } i=\ell \\ 0 & \text{if } i\neq\ell \end{cases}}$$

$$= \sum_{i=1}^{P} |a_i|^2 e^{j2\pi f_i k}$$

Form $\underline{x}[t] = \begin{bmatrix} x[t] \\ x[t+1] \\ \vdots \\ x[t+M-1] \end{bmatrix}$ ($M \times 1$ vector)

$$R_{xx} = E[\underline{x}[t]\, \underline{x}[t]^H] = E\left(\begin{bmatrix} x[t] \\ x[t+1] \\ \vdots \\ x[t+M-1] \end{bmatrix} \begin{bmatrix} x[t]^* & x[t+1]^* & \cdots & x[t+M-1]^* \end{bmatrix}\right)$$

$$= \begin{bmatrix} r_{xx}[0] & r_{xx}[-1] & \cdots & r_{xx}[-(M-1)] \\ r_{xx}[1] & r_{xx}[0] & \cdots & r_{xx}[-(M-2)] \\ \vdots & & & \\ r_{xx}[M-1] & r_{xx}[M-2] & \cdots & r_{xx}[0] \end{bmatrix}$$

$$R_{xx}^H = R_{xx}$$

$$R_{xx} = \sum_{i=1}^{P} |a_i|^2 \underline{s}_i \underline{s}_i^H, \qquad \underline{s}_i = \begin{bmatrix} 1 \\ e^{j2\pi f_i} \\ e^{j2\pi 2 f_i} \\ \vdots \\ e^{j2\pi(M-1)f_i} \end{bmatrix}$$

$$S = [\underline{s}_1 \quad \underline{s}_2 \quad \cdots \quad \underline{s}_P]$$

$$R_{xx} = S P S^H, \qquad P = \begin{bmatrix} |a_1|^2 & & \\ & \ddots & \\ & & \end{bmatrix}$$

$$S = [\, \underline{s}_1 \quad \underline{s}_2 \quad \cdots \quad \underline{s}_p \,]$$

$$\underbrace{R_{xx} = S P S^H}$$

$$P = \begin{bmatrix} |q_1|^2 & & & \\ & |q_2|^2 & & \\ & & \ddots & \\ & & & |q_p|^2 \end{bmatrix}$$

If we do the eigen d

$$R_{x_1} = U \Lambda U^H$$

$$U = [\, \underline{u}_1 \; \underline{u}_2 \; \cdots \; \underline{u}_M \,]$$

eigen